



Beyond text generation: large language models as autonomous agents for orchestrating synthetic biology cycles

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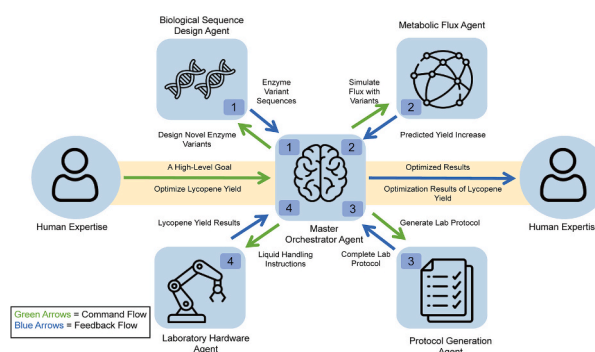
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HIGHLIGHTS

- Large language models (LLMs) optimize and accelerate the design-build-test-learn (DBTL) cycle in synthetic biology.
- LLMs shift from manual features to semantic-driven approaches, unlike traditional machine learning (ML).
- Specialized LLMs, combined with an orchestrator agent, advance the development of synthetic biology.

GRAPHICAL ABSTRACT



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ABSTRACT

Synthetic biology, which aims to design and construct novel biological systems, faces inherent challenges in navigating high-dimensional design spaces and optimizing complex experimental cycles. Large language models (LLMs) are emerging as semantic engines capable of parsing the deep logic of biological systems, making them particularly suitable for processing complex biological sequence data. In this review, we first systematically show how different LLM architectures are uniquely suited to specific biological tasks. We also highlight that LLMs shift the paradigm from manual feature creation to semantic-driven approaches, compared to traditional machine learning. Through a critical examination of transformative applications in novel biological sequence design, context-aware metabolic engineering, and intelligent automation laboratories, we chart both the unprecedented accelerations and the profound new challenges, including computational cost, interpretability, and ethical governance, that this paradigm introduces. We position LLMs not merely as predictive tools, but as autonomous agents that orchestrate the synthetic biology design-build-test-learn cycle. The emergence of a collaborative

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intelligence, where human expertise is dynamically partnered with multiple agents to navigate the vast design space of biology, steers the field toward breakthroughs in sustainable manufacturing, medicine, and beyond.

1. Introduction

Synthetic biology represents a concerted effort to engineer biological systems into programmable platforms. Its applications span critical domains including biotherapeutics (Munirah et al., 2025; Omid et al., 2024), sustainable biofuel production (Meijnikman et al., 2022), and environmental remediation (Gao et al., 2024). The foundational framework driving this field is the design-build-test-learn (DBTL) cycle, which applies engineering principles to biological design (Foldi et al., 2024; Lu et al., 2024). However, scaling laboratory demonstrations to industrial applications confronts the complexity inherent in biological systems (Qiao et al., 2025). This complexity creates three pervasive bottlenecks. First, knowledge fragmentation scatters crucial design rules across unstructured literature and disparate databases, hindering systems-level insight. Second, experimental inefficiency confronts a combinatorial explosion of possible biological variants, making exhaustive search impractical (Boro et al., 2025). Third, and most critically, a cognitive gap exists in automation. While capable of executing predefined protocols, they can not interpret experimental context or dynamically respond to unforeseen biological outcomes, restricting their utility in iterative design workflows (Barreiro et al., 2024). Consequently, even well-established modeling tools like flux balance analysis (FBA) often yield predictions that diverge from experimental results, largely due to unaccounted dynamic variables and environmental perturbations within living cells (Li et al., 2022; Sen, 2024).

Large language models (LLMs) are general-purpose artificial intelligence (AI) systems built on Transformer architectures, pre-trained with massive text data. Their core mechanism involves modeling deep correlations between linguistic symbols through self-attention mechanisms, enabling cross-task generalization from text understanding to generation. By leveraging self-attention mechanisms, LLMs model deep semantic relationships within biological knowledge. They transcend pattern recognition to enable reasoning across disparate concepts, synthesizing fragmented information, and translating high-level directives into actionable experimental plans (Zhang et al., 2025d; Zhou et al., 2025b). Furthermore, agents are systems that utilize LLMs as their cognitive core, capable of perceiving the environment, conducting logical reasoning, and autonomously operating tools to achieve specific scientific objectives (Ramos et al., 2024). In this review, we regard LLMs applied to various components of synthetic biology as specialized expert agents (e.g., biological sequence design agent, metabolic flux agent, and so on), with a master orchestrator agent that coordinates all expert agents to achieve global optimization.

As a cutting-edge AI technology, LLMs have demonstrated significant application potential across various research areas in synthetic biology in recent years, offering novel pathways to address the field's fundamental scientific challenges. In biological sequence design, LLMs leverage their ability to learn the deep grammar of life from massive databases, enabling the *de novo* generation of functional, diverse proteins and nucleic acids that transcend the limitations of natural evolutionary templates. For example, ProGen is an LLM-based protein design model that can generate novel protein sequences based on specific functional or structural requirements (Madani et al., 2023). In metabolic engineering, LLMs excel at integrating heterogeneous multi-omics data to provide context-aware predictions of enzyme kinetics and complex pathway dynamics, overcoming the accuracy limitations of traditional mechanistic modeling. A case in point is UniKP, it can accurately predict enzyme kinetic parameters based on LLMs (Yu et al., 2023). In intelligent automation laboratories, LLMs function as autonomous agents capable of translating high-level scientific directives into executable robotic protocols and interpreting experimental feedback to close the

DBTL cycle, reducing the high labor cost of experiments. Take SAMPLE as an example, it can design new proteins and send designs to a fully automated robotic system that experimentally tests the designed proteins (Rapp et al., 2024).

Previous reviews have cataloged machine learning (ML) applications within the DBTL cycle or detailed specific deep learning (DL) architectures for biological tasks. For example, Faulon et al. reviewed the application of supervised and semi-supervised ML in predicting enzyme activity and optimizing metabolic pathways (Faulon and Faure, 2021). Rai et al. argue that adopting data-driven approaches that integrate modern ML tools and high-throughput experimental approaches into the synthetic biology DBTL process could dramatically accelerate the pace and scope of bacterial design (Rai et al., 2024). Goshisht described predictive potential and basics of DL along with myriad applications in synthetic biology, especially in engineering cells, activity of proteins, and metabolic pathways (Goshisht, 2024). However, the technology they mentioned relies on manual curation and existing domain knowledge rather than autonomously discovering and encoding latent biophysical and evolutionary features.

This review provides a fresh perspective on the role of LLMs in synthetic biology. First, we introduce the main architectures of LLMs and delineate their appropriate roles within the biological domain, discussing the paradigm shift they represent compared to traditional ML. Second, we detail the specific applications of different LLMs across key areas: biological sequence design, metabolic engineering, and intelligent automation laboratories. We also provide real cases about them. Furthermore, we examine current challenges, such as high computational cost, interpretability, hallucination, and areas where LLMs still underperform conventional ML. Finally, we conclude that the future lies in specialized multi-agent systems, where autonomous domain-specific agents leverage their expertise through optimized, orchestrated collaboration to achieve superior global objectives.

2. Main architectures of LLMs

The transformative potential of LLMs in synthetic biology is not merely a function of their scale but a direct consequence of their underlying computational architectures. These architectures impose specific priors, or biases in how information is processed, which fundamentally reshape how we can represent, interrogate, and generate biological data. In this section, we examine how core LLM paradigms differentially equip researchers to tackle the distinct representational challenges of biology.

2.1. Autoregressive model

Autoregressive models, epitomized by the GPT lineage, operationalize a simple yet powerful constraint: generate the next token based strictly on what came before (Barbieri and Diversi, 2024; Kikilishvili, 2024). Formally, it models the probability distribution $P(X)$. Implemented via causally masked Transformer decoders, this architecture excels at *de novo* sequence generation (Fig. 1A). For synthetic biology, this prior is uniquely suited to tasks where the output is itself a linear string, such as designing novel protein sequences, synthetic promoters, or guide RNA libraries. The model learns the “grammar” of biological sequences from vast corpora and can then sample new, syntactically valid instances.

$$P(X) = \prod_{t=1}^n P(x_t | x_{<t}), \quad (1)$$

Critically, this strength is also a limitation. Its logic is fundamentally

additive and forward-moving. The unidirectional information flow creates myopia for the global context. While excellent for inventing a novel enzyme sequence, a purely autoregressive model may struggle with tasks requiring holistic, non-sequential reasoning. This includes predicting how a newly generated protein fold will interact with a suite of metabolites.

2.2. Autoencoder model

Autoencoder-style models like BERT and its protein-specific descendants employ a bidirectional attention mechanism (Acheampong et al., 2021; Zhou et al., 2025a). Their pre-training objective, which is often masked language modeling, forces the model to build a deep, contextualized representation of every element in a sequence (Fig. 1B). In this approach, random tokens are obscured and must be predicted from their full context (Rethmeier and Augenstein, 2023; Usuyama et al., 2025).

This architectural prior is ideal for interpretation and prediction. By seeing the entire sequence at once, the model captures long-range dependencies and co-evolutionary signals that often correlate with 3D structure and function. Consequently, these models shine at tasks like predicting the effect of point mutations, annotating protein function, or identifying structurally critical residues. The evolution from general-purpose BERT to large-scale protein language models demonstrates that increasing depth directly enhances the model's ability to infer spatial and functional relationships from primary sequence data. While primarily serving as robust analytical engines for inference, these models also provide the structural priors necessary for inverse folding and scaffold-constrained design.

2.3. Encoder-decoder architecture

The encoder-decoder architecture introduces a separation of concerns: comprehend an input, then generate an output (Fig. 1C). Frameworks like T5 formalize diverse tasks as “text-to-text” translation (Hirschberg and Manning, 2015; Raffel et al., 2020). In a biological context, this becomes a powerful paradigm for cross-modal translation. The input could be a natural language description of a desired phenotype, and the output could be a protein sequence. Alternatively, it could translate a metabolic network graph into a simulation report or a noisy experimental readout into a diagnostic conclusion.

This architecture is the backbone for building agentic systems. The encoder can process multimodal input, building a latent representation that the decoder uses to generate actionable output, such as a robot-compatible experimental protocol or a hypothesis. Emerging hybrids, like models employing mixture-of-experts (MoE) layers or prefix-tuning, are optimizing this framework for efficiency, enabling it to handle the scalability demands of combinatorial library design or multi-omics data integration (Fedus et al., 2022; Shaik et al., 2022).

2.4. The paradigm shift from manual feature creation to semantic-driven approaches

Positioning LLMs against traditional ML underscores a fundamental paradigm shift in how biological intelligence is engineered (Fig. 2). Classical ML methods, including support vector machines (SVMs) and random forests, operate within a feature-constrained paradigm. Their performance is inherently bounded by expert-defined feature sets, such as hydrophobicity indices, solubility scores, or structural descriptors, which require substantial domain knowledge and manual curation (Jordan and Mitchell, 2015). While effective for well-characterized tasks, this approach struggles with the complexity, high-dimensionality, and contextual interdependencies inherent in biological systems. For instance, in k_{cat} prediction tasks, UniKP, which utilizes an LLM, achieves an average coefficient of determination (R^2) that is 20% higher than that of the traditional DL model DLKcat (Yu et al., 2023).

In contrast, LLMs represent a qualitative leap (Table 1), offering three core advantages that redefine computational approaches to synthetic biology:

Automated latent representation learning: Through self-supervised pre-training on vast, diverse biological corpora, LLMs autonomously discover and encode latent features that capture complex biophysical, evolutionary, and functional relationships. This moves the research bottleneck from “how do we manually describe this biological problem?” to “how do we properly task the model?”, which enables the model to uncover representations that often surpass human-engineered descriptors in predictive power and generalizability.

Cross-domain semantic reasoning: Unlike traditional ML models that excel within narrow, feature-defined task boundaries, LLMs leverage transformer-based attention mechanisms to build a unified semantic understanding across disparate biological concepts. This allows them to reason across modalities in a manner that mirrors integrative scientific reasoning, for instance, linking gene sequence, protein structure, metabolic function, and relevant literature evidence. This capability for context-aware synthesis is crucial for tasks such as designing functional protein complexes or engineering metabolic pathways where multiple constraints interact.

Few-shot and zero-shot generalization: LLMs can make meaningful predictions for novel protein families, pathways, or organisms with minimal to no task-specific labeled data (Cheng et al., 2024). This is particularly transformative in data-sparse domains of synthetic biology, where exhaustive experimental labeling is impractical. By leveraging patterns learned during pre-training, LLMs can infer function, optimize sequences, or predict interactions for targets with no close homologs, dramatically accelerating exploration of biological “dark matter.”

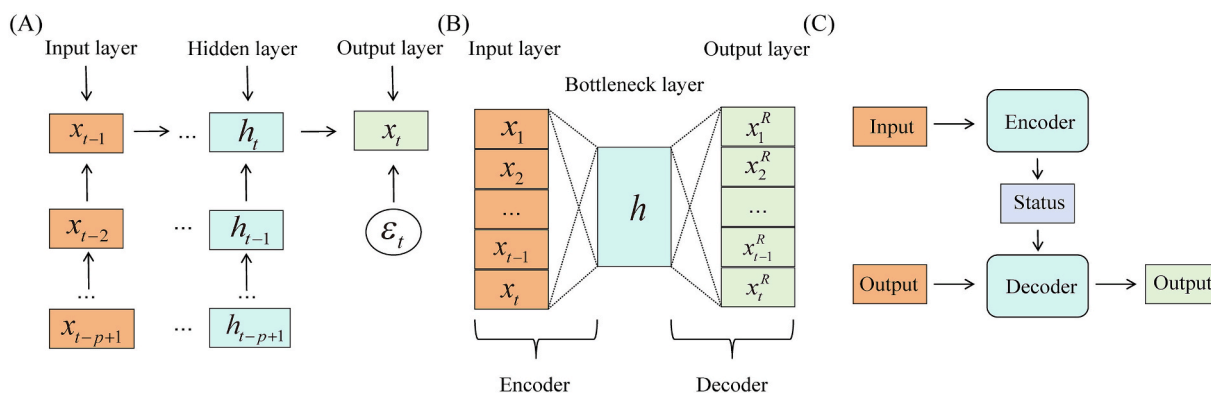


Fig. 1. The main categories of the major language models. (A) Autoregressive model; (B) Autoencoder model; (C) Encoder-decoder architecture.

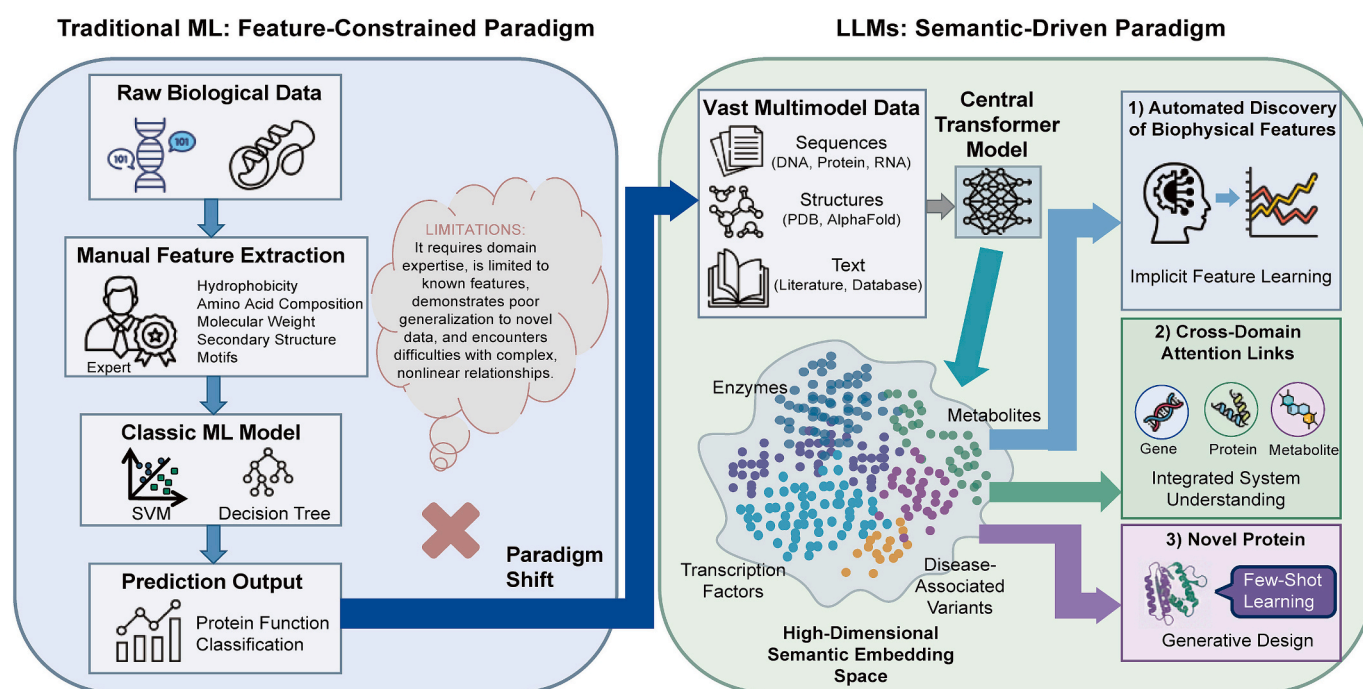


Fig. 2. Paradigm shift from traditional ML to LLMs.

Table 1

Dimensional contrasts between traditional ML and LLMs.

Dimension	Traditional ML	LLMs
Feature engineering	Rely on expert-defined feature sets (e.g., hydrophobicity, molecular weight). Performance is bounded by the quality of manual curation and existing domain knowledge.	Autonomously discover and encode latent biophysical and evolutionary features through self-supervised pre-training on massive biological corpora.
Data needs	Typically require large, curated datasets for specific tasks. Often struggle with data sparsity where exhaustive experimental labeling is impractical.	Be able to make meaningful predictions for novel proteins or pathways with minimal to no task-specific labeled data by leveraging pre-learned patterns.
Reasoning scope	Operate within fixed, feature-defined boundaries. Often struggle with cross-domain interdependencies (e.g., linking text to structure).	Use self-attention mechanisms to build a unified understanding across disparate concepts (e.g., linking gene sequence, metabolic function, and literature).
Typical use cases	Work best for well-characterized tasks like protein classification or predicting specific kinetic parameters based on known descriptors.	Aim to tackle the <i>de novo</i> design of novel sequences, explore the protein “dark matter”, and integrate multi-omics data for holistic, system-level design.

This shift does not come without trade-offs. The computational intensity of training and inference, model opacity, and reliance on large-scale data represent significant challenges (Zhuang et al., 2024). However, advances in efficient fine-tuning, model distillation, and retrieval-augmented generation (RAG) are rapidly making these powerful models more accessible and grounded for real-world biological design (Ye et al., 2025).

In summary, LLMs are not merely incremental improvements over traditional ML; they represent a different class of tools that move from pattern recognition within fixed feature spaces to context-aware reasoning and generation within a learned semantic space. This makes them uniquely suited to navigate the complexity and uncertainty inherent in engineering living systems.

3. Biological sequence design with LLMs

The design of biological sequences is undergoing a decisive transition. We are moving beyond local, experience-guided mutations toward a paradigm of systematic navigation within a high-dimensional fitness landscape, powered by models that treat biomacromolecules as a formal language with learnable syntax and semantics. We collect LLMs applied to synthetic biology and analyze the differences among them (Table 2).

3.1. From description to generation

The journey began with solving a critical inverse problem: sequence to structure. AlphaFold2's transformer-based architecture demonstrated that self-attention could capture long-range residue couplings to predict 3D folds with remarkable accuracy (Olanders et al., 2024; Yang et al., 2023). RGN2 and ESMFold both leverage transformer-based large language models, trained on extensive protein sequence data, to predict three-dimensional protein structures directly from amino acid sequences, by utilizing deep contextual representations learned during pre-training, thereby bypassing the need for the computationally expensive multiple sequence alignment (MSA) step that is central to AlphaFold2's approach. (Chowdhury et al., 2022; Lin et al., 2023). However, these are primarily descriptive tools.

The frontier has now shifted decisively to generation. Models like ProGen exemplify this leap. Trained in a controlled, label-conditioned manner on 280 million sequences across 19,000 families, it learns a generative grammar of proteins (Madani et al., 2023). This enables function-first design: by specifying desired properties, ProGen can generate novel, stable protein sequences that fold into functional structures, as validated by subsequent wet-lab characterization of synthetic libraries (Lee et al., 2025). ProGen2, scaled to over one billion sequences, further advanced the state-of-the-art in generating viable,

Table 2
LLMs and models based on LLMs applied to synthetic biology.

Name	Application	Base models	Key architecture type	Advantages	Limitations	Availability	Functional classification
RGN2 (Chowdhury et al.,2022)	Prediction of the structures of orphan proteins (those without homologous sequences) and <i>de novo</i> designed proteins	AminoBERT	Encoder-Decoder	MSA-free prediction from a single sequence enables orders-of-magnitude speed increases compared to MSA-dependent methods	Weak ability to predict β -folding structures	https://github.com/aqlaboratory/rgn2	Structure prediction
ESMFold (Lin et al.,2023)	Directly extracting implicit structural and evolutionary information from a single sequence	ESM-2	Encoder-Decoder	Fast computing speed and multi-dimensional information extraction	Overall accuracy is slightly lower, and the prediction of some special structures is not good	https://github.com/facebookresearch/esm	Structure prediction
ProGen (Madani et al.,2023)	Generation of novel proteins with specific binding activity or catalytic function	Transformer Decoder	Autoregressive	Powerful sequence generation capability, covering a wide range of protein spaces	Lack of direct modeling of structure and function raises doubts about the feasibility of generating sequences	https://github.com/salesforce/progen	Molecular generation
ProGen2 (Nijkamp et al.,2023)	Generation of precise new proteins with specific binding activity or catalytic function	Transformer Decoder	Autoregressive	The rationality of the structure and function of the generated sequence has been significantly improved	The computational cost has significantly increased, the threshold for use has been raised, and the deep mechanism modeling of structure and function is still limited	https://github.com/enijkamp/progen2	Molecular generation
ESM-2 (Lin et al.,2023)	The use of biological priors to guide directed evolution or active learning loops	ESM-2	Autoencoder	Powerful sequence representation capability, excellent downstream task transferability, adaptability to different resource requirements, and strong interpretability	Lack of explicit structure and physical constraint modeling, and weak generation ability	https://github.com/facebookresearch/esm	Sequence representation
ProtGPT2 (Ferruz et al.,2022)	Generation of proteins that exhibit natural amino acid tendencies	GPT-2	Autoregressive	The naturalness of the generated sequence is high, and its statistical characteristics are close to real proteins. It can generate novel but reasonable sequences and expand the protein space	The model size and performance are limited, and there is a gap compared to the new generation model	https://github.com/TeletcheaLab/protGPT2	Molecular generation
SYMPLEX (Wang et al.,2025a)	Extraction and analysis of literature content and the subsequent generation of a high-quality candidate gene set with a complete chain of evidence	BioBERT	Encoder-Decoder	Cross-modal knowledge integration breaks through the limitations of single data, and automating literature analysis significantly improves mining efficiency	Strong dependence on literature quality, and a high threshold for computing resources and domain knowledge	https://bdainformatics.org/page?type=SYMPLEX	Molecular generation
Evo (Nguyen et al.,2024)	Molecular to genome-scale sequence modeling and design of prokaryotes and bacteriophages	StripedHyena	Autoregressive	Strong cross-modal prediction ability, capable of generating and designing complex biological systems	The modeling capability of eukaryotic organisms is limited, and the demand for computing resources is high	https://github.com/evo-design/evo	Molecular generation
Evo2 (Brixi et al.,2025)	Molecular to genome-scale sequence modeling and design of bacteria, archaea, eukaryotes, and bacteriophages	StripedHyena 2	Autoregressive	Abundant coverage of prokaryotic and eukaryotic data, and strong interpretability	Insufficient modeling of biological regulatory mechanisms and a high false positive rate in predicting promoter enhancer interactions	https://github.com/arcinstitute/evo2	Molecular generation
DNABERT (Ji et al.,2021)	Prediction of gene expression levels and optimization of codon usage preferences	BERT	Autoencoder	Outstanding transfer learning ability and strong interpretability	Limited ability to model long-range dependencies and insufficient species generalization	https://github.com/jerryji1993/DNABERT	Function prediction
EVOLVEpro (Jiang et al.,2025b)	Protein directed evolution and functional optimization	ESM-2	Autoregressive	Active learning with “few samples”, significant improvement in protein function can be achieved with only a few rounds of experiments	The ability relies on protein language models and still requires wet experimental validation	https://github.com/mat10d/evolvepro	Function prediction
ESM-Ezy (Qian et al.,2025a)	Screening of copper oxidase	ESM-2	Autoencoder	Breaking through the limitations of sequence similarity and efficiently discovering new enzymes	Relying on high-quality annotated data limits generalization ability	https://github.com/westlake-rep1/ESM-Ezy	Function prediction
UniKP (Yu et al.,2023)	Prediction of k_{cat} , K_M , and k_{cat}/K_M	ProtT5	Encoder-Decoder	Environmental factors and high-value prediction optimization, lightweight and efficient	Insufficient robustness to extreme environments, and difficulty in model interpretability and mechanism analysis	https://github.com/Luo-SynBioLab/UniKP	Function prediction
EPHOD (Gado et al.,2025)	Prediction of optimal pH value	ESM-1	Encoder-Decoder	The attention weights of the model can locate residues adjacent to the catalytic center, revealing key determinants of pH	High computational cost and model complexity, and limited prediction for extreme pH conditions	https://github.com/beckham-lab/EpHod	Function prediction

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Table 2 (continued)

Name	Application	Base models	Key architecture type	Advantages	Limitations	Availability	Functional classification
CAR-Toner (Qiu et al.,2024)	Optimization of CAR-T cell design	ESM-2	Encoder-Decoder	High precision prediction, convenient and fast calculation	Multi-objective optimization is difficult, and off-target effects may occur	https://cart-fitness.sls.t.shanghaitech.edu.cn/CAR-fitness/	Function prediction
D2Cell (Li et al.,2024)	Method of data collection in metabolic engineering	GPT-4	Autoencoder	Automatically extract metabolic engineering knowledge from existing literature to support applications in the field of metabolic engineering	Limitations of databases and large computing resources	https://github.com/LiLabTsinghua/D2Cell	Data extraction
Prot2Chat (Wang et al.,2025b)	Understanding the sequence and structure of proteins, and answering questions about their function, localization, structural domains, etc. in natural language	Llama	Encoder-Decoder	Unified fusion of sequence and 3D structural information at the model input to achieve more accurate cross modal understanding capability	The interpretability of features is limited, and existing training data may be biased towards specific species, affecting generalization ability	https://github.com/wangzcl1233/Prot2Chat	Data fusion
ProtTeX (Ma et al.,2025)	Unified processing of tasks such as functional prediction, structural conformation generation, and text-based protein design	—	Autoregressive	The joint representation and training of protein sequences, 3D structures, and textual information in the same discrete space significantly enhances the model's ability to understand and generate across modalities	The unified dense representation and joint training process require high computational resources	https://github.com/mzc2113391/ProtTeX	Data fusion
SynBioGPT v2.0 (Mao et al.,2025)	A platform to guide the development of microbial strains	Llama	Autoregressive	Outstanding performance in core tasks of synthetic biology, and high precision in understanding professional terminology	Insufficient support for rare species or new technologies, weak global optimization capability for complex systems, and poor interpretability	https://synbiogpt.bio-design.ac.cn	Automated design
ChemCrow (Bran et al.,2024)	Generation platform from target molecular structure formula to complete synthesis route	GPT-4	Autoregressive	Cross-tool collaboration and task automation, domain knowledge enhancement, and accuracy improvement	The closed-source feature of GPT-4 makes it difficult to deeply optimize inference logic, and the calling cost is relatively high	https://github.com/ur-whitelab/chemcrow-public	Automated design
PLMeAE (Zhang et al.,2025b)	A platform for automated construction and testing of the target protein	ESM-2	Autoencoder	The experimental period has been significantly shortened, and it is not only suitable for optimizing natural proteins, but also for enhancing non-natural amino acids	Insufficient data on non-natural substrates, and high experimental dependence and cost threshold	https://github.com/HICAI-ZJU/PLMeAE_Project	Automated design
Biomni (Huang et al.,2025)	Biomedical AI agent	GPT-4	Autoregressive	Strong cross-domain task generalization ability, user-friendly, and open-source ecosystem	Closed-source model limitations and security risks	https://github.com/snap-stanford/Biomni	Automated design
SAMPLE (Rapp et al.,2024)	Self-driving laboratories to autonomously navigate the protein fitness landscape	—	—	Breaking through manual dependence, achieving a DBTL closed-loop without human intervention, and high repeatability of experiments: standardized processes reduce human errors	The size and complexity of the sequence space are limited; the range of functional prediction is narrow, which relies on specific detection methods; the threshold for equipment and maintenance is high	https://doi.org/10.5281/zenodo.10048592	Automated design

active proteins without requiring fine-tuning (Nijkamp et al., 2023). These models act as learned priors over evolutionary space, enabling the creation of sequences that are both novel and functionally plausible.

3.2. Navigating the semantic landscape

The generative capability hinges on a core innovation: the projection of discrete sequences into a continuous, high-dimensional semantic embedding space. The ESM-2 protein language model, through self-supervised learning on billions of residues, creates embeddings that encode biophysical properties and evolutionary constraints (Lin et al., 2023). This latent space is operationally powerful. It enabled the construction of the ESM Metagenomic Atlas, which predicted structures for over 617 million metagenomic proteins, vastly expanding the map of known protein fold space and revealing functional dark matter (Zhang et al., 2025a). Crucially, this representation allows for semantic interpolation and functional prediction for sequences with no close homologs, addressing the endemic data sparsity problem. ESM-Ezy leverages ESM embeddings combined with structural features to screen metagenomic data for novel polycopper oxidases, successfully identifying high-activity, low-homology enzymes for bioremediation (Qian et al., 2025a).

3.3. Scaling design logic across molecular and genomic systems

Design ambition is expanding from single molecules to multi-component systems. For genetic part optimization, models like DNA-BERT analyze nucleotide-level patterns in promoters and terminators to predict and optimize gene expression levels and codon usage, providing a fine-tuning tool for synthetic constructs (Consens et al., 2025; Ji et al., 2021).

The most profound scaling is seen in whole-genome language models. Evo, a 7B-parameter model based on the StripedHyena architecture trained on 2.7 million prokaryotic and phage genomes, can perform both sequence generation and zero-shot function prediction at genomic scales (Nguyen et al., 2024). Its successor, Evo2, trained on a staggering 9.3 trillion DNA bases, learns complex features across all domains of life, enabling the *in silico* design of functional systems like CRISPR-Cas arrays and transposons, which have been experimentally validated (Brixi et al., 2025). This marks the transition from designing parts to designing systems and genomic contexts.

4. Metabolic engineering with LLMs

The engineering of cellular metabolism represents one of synthetic biology's most formidable challenges, which is essentially a multidimensional optimization puzzle where genetic perturbations, enzyme kinetics, and environmental fluctuations interact non-linearly. Traditional tools, often confined to steady-state abstractions or manual literature curation, struggle to navigate this complexity. LLMs are emerging as crucial integrative reasoners, translating between biological knowledge, mechanistic models, and engineering goals to enable dynamic, predictive design.

4.1. Dynamic knowledge integration

Metabolic engineering has long been bottlenecked by knowledge fragmentation. Valuable design rules, like which genes to knock out or which heterologous pathways to express, are buried in unstructured text. LLMs are transforming this landscape by automating the construction of actionable knowledge bases. The D2Cell database exemplifies this shift (Li et al., 2024). By processing vast volumes of literature, LLMs extracted and structured over 29,000 metabolic engineering records, linking products, organisms, and strategies into a computable resource. This model demonstrated superior generalization in predicting engineering targets, even for novel products, moving beyond the limits

of manually curated pathways.

This capability extends to cross-modal alignment, a critical need in bioinformatics. Emerging approaches such as Prot2Chat and ProtTeX illustrate how LLMs can bridge the semantic gap between protein representations and natural language without domain-specific retraining. Prot2Chat functions as a novel question–answer framework that integrates both protein sequence and structural information, enabling interactive, context-aware dialogue about protein function and design (Wang et al., 2025b). Meanwhile, ProtTeX advances further by not only tokenizing amino acid sequences but also integrating 3D structural data and relevant textual descriptions into a unified discrete latent space (Ma et al., 2025). This allows the model to dynamically leverage multimodal information during inference, achieving state-of-the-art performance in protein-to-text and text-to-protein generation tasks.

4.2. Precise kinetic prediction

A core weakness of conventional metabolic modeling, such as FBA, is its reliance on the steady-state assumption and generic kinetic parameters. This overlooks the enzyme-specific reality of catalysis, governed by constants like k_{cat} and K_m , which are sparse and context-dependent. LLMs are now providing a path to high-fidelity kinetic prediction.

Models like UniKP tackle this directly (Yu et al., 2023). By integrating a protein language model (ProtT5) for enzyme sequence encoding with a SMILES transformer for substrate structure (Yang et al., 2026; Zhang et al., 2026), UniKP predicts kinetic parameters directly from sequence and substrate information. Its successor, EF-UniKP, goes further by incorporating environmental factors (pH, temperature) through an ensemble framework. Similarly, EPHOD uses protein sequence embeddings from ESM-1 to predict an enzyme's optimal pH (pHopt), revealing residue-level features critical for activity under different conditions (Gado et al., 2025). These tools make it possible to transition metabolism from a stoichiometric network to a kinetically informed, dynamic system.

4.3. Intelligent metabolic agents

The ultimate goal is not just to model metabolism, but to allow it to self-optimize within a design cycle. This requires transitioning from static simulations to interactive, agent-based frameworks. The SPAM-DFBA (Self-Playing Microbes in Dynamic FBA) algorithm represents this frontier (Ghadermazi and Chan, 2024). It treats each microbial strain as a reinforcement learning agent that learns to regulate its metabolic fluxes in a dynamic environment, balancing immediate growth against long-term survival strategies, such as predicting enzyme secretion in nutrient gradients. This transcends the traditional “biomass maximization” of classic FBA (Ye et al., 2024).

Integrating this agentive reasoning into the DBTL cycle is the function of platforms like SynBioGPT v2.0 (Mao et al., 2025). Acting as a central reasoning engine, it mitigates the hallucination and knowledge recency problems of general-purpose LLMs through RAG. It can synthesize multi-omics data, literature evidence, and mechanistic model outputs to propose optimal strain designs, effectively orchestrating the iterative “Learn” and “Design” phases. Here, LLMs become the strategic planner, dynamically navigating the metabolic fitness landscape based on continuous feedback.

The convergence of several key capabilities points toward a new paradigm: intelligent metabolic agents. These capabilities encompass automated knowledge extraction, precise kinetic prediction, and agent-based optimization. In this vision, an LLM-powered agent interprets a high-level production target, mines the literature for relevant strategies, designs a tailored kinetic model to simulate outcomes, proposes a prioritized set of genetic edits, and iteratively refines its proposals based on experimental omics data from bioreactors. The challenge shifts from gathering enough data to building reasoning systems capable of interpreting and acting upon it. The promise is a future where metabolic

networks are not just constructed but continuously evolved *in silico* and deployed *in vivo* with unprecedented speed and precision (Guo et al., 2025; Juanpere-Borras et al., 2026).

5. Intelligent automation laboratories with LLMs

The laboratory bench is witnessing a silent revolution. For decades, automation meant one thing: the meticulous translation of human intent into brittle, linear code, that is, scripts that could move liquids but couldn't comprehend an experiment. This paradigm is shattering. LLMs are injecting semantic understanding into the very machinery of science, evolving tools from executors into collaborators. The shift isn't merely technical; it's epistemological, transforming how we conceive of and conduct biological experimentation.

5.1. Bridging the semantic gap in experimentation

Traditional laboratory automation operates based on predefined protocols, executing procedures with precision but without comprehending their scientific intent. For instance, a pipetting robot follows programmed instructions specifying exact volumes and coordinates, yet remains unaware of the higher-level purpose of the transfer, such as initiating a phosphorylation cascade essential to testing a specific hypothesis. This semantic gap between human reasoning and machine execution has long constrained the development of high-level, adaptive experimentation.

LLMs are building the bridge. They achieve this by deconstructing the experimental protocol, a narrative document, into a structured, actionable knowledge graph. Consider ChemCrow (Bran et al., 2024). When tasked with synthesizing ibuprofen, it doesn't retrieve a static recipe. Instead, it reasons: parsing the SMILES string, accessing databases of organic reactions, and assembling a coherent, step-by-step synthetic pathway complete with stoichiometry, hazard warnings, and necessary equipment. The agent evaluates the requirements of a target molecule and independently selects the appropriate tools to assemble an actionable protocol. It understands that "add dropwise" implies a need for temperature control and a specific type of condenser. This is not script generation; it is procedural reasoning, transforming a goal into a logically sound, executable plan. Similarly, systems leveraging ChatGPT for Robotics demonstrate how natural language instructions are dynamically decomposed into low-level robot commands, spatial planning for collision avoidance, and real-time error correction routines (Vemprala et al., 2024; Zhang et al., 2025c).

5.2. Efficient autonomous experimentation

Self-driving laboratories (SDLs) are AI-powered platforms with a self-accelerating nature, using robotics and other hardware to automate experiments and data collection (Delgado-Licona et al., 2025). Here, LLMs act as the proposal engine within an active learning or reinforcement learning framework. A canonical example is the optimization of antibodies against the SARS-CoV-2 spike protein: model-generated variant libraries were screened, and wet-lab validation data were fed back to refine the model, leading to highly potent neutralizing antibodies with far greater efficiency than random mutagenesis (Dauparas et al., 2022; Thadani et al., 2023). This iterative three-phase cycle, comprising *in silico* proposal, experimental testing, and model refinement, constitutes the operational core of the emerging SDLs (Desai et al., 2025; Liu and Kalinin, 2025). Platforms like SYMPLEX extend this logic upstream, using LLMs to mine millions of documents to extract candidate genes with full evidence chains, thereby priming the design pipeline with high-quality, knowledge-anchored starting points (Wang et al., 2025a).

The operation of SDLs constitutes a closed loop:

- **Design:** The LLM agent, given a goal, navigates the protein fitness landscape. It doesn't just generate random mutations; it uses its embedded knowledge of structure–function relationships to propose intelligent, high-likelihood variants.
- **Build/Test:** It then authors the experiment to test itself. It generates the robotic code to synthesize DNA, express proteins, and assay activity, orchestrating the physical hardware of the biofactory.
- **Learn:** Crucially, it ingests the results, namely the raw data on growth, expression, and enzymatic function. Through what can be termed computational self-reflection, it analyzes failure and success, updating its internal model to refine the next design cycle. This transforms the DBTL cycle from a human-managed process with inherent delays into a compressed, autonomous discovery engine. The proof is in the outcomes: such systems have autonomously evolved enzymes with enhanced properties in a fraction of the traditional timeline.

5.3. The multi-agent collaborative ecosystem

The frontier lies in moving beyond a single, monolithic AI. The complexity of biological systems necessitates modular, specialized architectures. In this framework, specialized agents, such as those dedicated to protein folding or metabolic flux, provide discrete functional insights that are integrated and prioritized by a master orchestrator agent. The orchestrator agent synthesizes these expert opinions, evaluates trade-offs between conflicting objectives, and resolves disagreements through mechanisms such as weighted voting, Bayesian optimization, or querying an external knowledge base via retrieval-augmented generation. If consensus remains elusive, the orchestrator may defer to human oversight for high-stakes decisions.

Platforms like CAR-Toner, which integrates structural biology and DL to optimize therapeutic cell design (Qiu et al., 2024), hint at this collaborative future. Generalist AI assistants like Biomni (Huang et al., 2025) could evolve into the interface for such a system, allowing a researcher to simply pose a challenge and watch as the agentic ecosystem formulates and executes a multi-year research plan. Biomni can autonomously identify, install, and call disparate bioinformatic tools to resolve multi-step queries. In the context of the "Design" and "Learn" phases of the DBTL cycle, Biomni is able to integrate outputs from literature-mining and computational tools to refine biological hypotheses. This is not mere automation; it is the augmentation of scientific intuition and reasoning, distributing cognitive labor across a hybrid human–machine collective.

Future laboratories are evolving into integrated, autonomous systems where the primary output extends beyond raw data acquisition to include the autonomous generation and iterative refinement of scientific hypotheses.

6. Real case study

The transition of LLMs from theoretical models to discovery partners is best illustrated through their deployment in orchestrating specific components of the multi-agent architecture across the DBTL cycle. In the following cases, we explicitly show how specific platforms function within our proposed framework. EVOLVEpro represents the sequence design agent, closing the loop between the "Learn" and "Design" phases through active learning. The integration of UniKP within the COBRA-k framework demonstrates how a metabolic agent enhances the "Design" phase with mechanistic grounding. Finally, PLMeAE exemplifies the laboratory hardware agent, where autonomous tool-calling and robotic orchestration drive the "Build" and "Test" stages toward physical realization.

6.1. Accelerated protein evolution

While traditional directed evolution typically requires exhaustive

and time-consuming experimental screening, EVOLVEpro overcomes this bottleneck by combining an ESM2-based feature extraction layer with a random forest activity predictor within an active learning loop (Jiang et al., 2025b). The underlying mechanism leverages ESM-2 semantic embeddings to capture essential biophysical priors and evolutionary constraints, enabling efficient navigation across the high-dimensional protein fitness landscape. As a result, this approach attains substantial functional improvement within only a few cycles of experimental feedback, exhibiting a strong few-shot capacity that greatly accelerates protein engineering. By using EVOLVEpro, the researchers successfully improved the binding affinity of two monoclonal antibodies by up to 40-fold, the indel formation activity of a miniature CRISPR nuclease by fivefold, showing the enormous power of combining LLMs to modify proteins.

6.2. Better metabolic network models

Traditional constraint-based reconstruction and analysis (COBRA) is a core method in systems biology for studying metabolic networks. Although valuable, this model is inherently static, typically assuming cells are in metabolic homeostasis. This approach overlooks the quantitative relationships between fluxes, enzyme abundance, and metabolite concentrations during dynamic changes. The COBRA-k model systematically integrates traditional constraint-based modeling with kinetic metabolic modeling (Bekiaris and Klamt, 2026). For enzyme kinetic parameters, accurate values can be obtained using LLM-based predictive tools such as UniKP. Researchers constructed and analyzed a large-scale COBRA-k model of central metabolism in *Escherichia coli*, validating the effectiveness and rationality of the COBRA-k approach. By setting biomass maximization as the objective, they found that the identified solution showed high consistency with metabolomic, proteomic, and phenomic data reflecting *E. coli* phenotypes, enabling more accurate metabolic network modeling.

6.3. General SDLs

Traditional protein optimization relies on multiple rounds of the DBTL cycle, with each round requiring manual operations, taking weeks to months to complete, and incurring high costs for reagents and equipment. Moreover, high-quality, standardized, and reproducible protein function data are scarce. In contrast, the SDLs ensure standardized and reproducible experimental data through fully automated workflows, while simultaneously monitoring multiple functional indicators. PLMeAE implements a dual-module architecture that leverages LLMs for protein variant design (Zhang et al., 2025b): the first predicts high-fitness mutations at unknown sites, and the second optimizes variants at pre-identified sites. This framework is supported by a robotic system that enables rapid construction of protein variants and continuous acquisition of functional data. Throughout the process, comprehensive metadata tracking and real-time data sharing maintain high reproducibility. Researchers validated this approach using the *p*-cyanophenylalanine tRNA synthetase from *Methanocaldococcus jannaschii* as a model enzyme. In each round of screening, they designed 96 variants using either LLMs or supervised ML models, which were then constructed and tested via an automated biofoundry platform. Across four rounds completed within two weeks, enzyme activity increased progressively, reaching its peak in the final round. These results demonstrate the effectiveness of the self-driving laboratory framework in accelerating protein engineering.

7. Current challenges and future directions

The ascent of LLMs from textual assistants to potential architects of biological systems is not a foregone conclusion. It is contingent upon solving a suite of profound, interconnected challenges that currently stand as gatekeepers between demonstration and deployment. These are

not mere technical drawbacks but fundamental questions about the reliability, efficiency, and safety of deploying probabilistic, data-driven intelligence at the heart of biological engineering (Fig. 3).

7.1. Computational cost and accessibility

The breathtaking performance of modern LLMs is built upon an equally breathtaking computational footprint. Training models like ProtGPT2 required weeks on clusters of high-end GPUs (Ferruz et al., 2022), while inference with large foundational models demands significant resources. This creates a democratization barrier, centralizing advanced capabilities in well-funded institutions and hindering widespread, iterative use in experimental settings.

The path forward lies in efficiency-aware architectural innovation. Techniques like low-rank adaptation (LoRA) for parameter-efficient fine-tuning, model quantization to reduce precision without catastrophic performance loss, and sparse MoE architectures that activate only relevant model pathways for a given task are critical. The goal is “foundation model” power with “specialist model” agility, enabling a small lab to fine-tune a massive protein language model on its proprietary strain data using a single workstation. Knowledge distillation, as seen in frameworks like SynBioGPT v2.0 (Mao et al., 2025), will further be key to compressing expert knowledge into leaner, more deployable models.

7.2. Interpretability and mechanistic grounding

In synthetic biology, a wrong prediction is not just an error; it can represent months of wasted experimental effort. The opaque decision-making of LLMs, manifesting in questions such as why it proposed a certain promoter variant or predicted a given metabolic flux, is therefore a major adoption barrier (Altorfer et al., 2025; Xiang et al., 2025). While attention maps offer clues about which parts of an input sequence the model focused on (Zhang et al., 2024), they do not explain the underlying biochemical rationale.

Future progress demands moving beyond visualization toward mechanistic grounding. This involves hybrid modeling, integrating LLMs with white-box, physics-based simulations. For instance, an LLM’s protein design suggestions could be constrained by or regularly validated against molecular dynamics simulations of folding stability. Physics-informed neural networks that embed known differential equations into the learning process can guide models toward physically plausible solutions. The objective is to build models that not only predict but also explain their predictions in the language of biochemistry.

7.3. Hallucination, bias, and data integrity

LLMs are fundamentally probabilistic generators, prone to producing confident yet fabricated outputs—“hallucinations.” (Farquhar et al., 2024) In a biological context, a hallucination could be a non-functional protein sequence, a toxic metabolic byproduct, or a falsely cited piece of literature. This unreliability is amplified by inherent biases in training data, which over-represent well-studied model organisms and proteins, leaving a “dark matter” of biological diversity poorly modeled (Mehrabian et al., 2021).

Combating this requires a multi-layered approach. RAG is paramount. By tethering the LLM’s reasoning to a curated, up-to-date knowledge base of verified biological facts, we can ground its outputs in evidence, as seen in agent platforms (Huang et al., 2025). Adversarial validation techniques, where a second model tries to distinguish real from generated biological data, can help identify and correct systematic biases (Lan et al., 2024; Wang et al., 2023). Finally, human-in-the-loop oversight remains a non-negotiable safety layer for high-stakes design tasks, ensuring a biologist’s expert intuition can vet and correct model proposals.

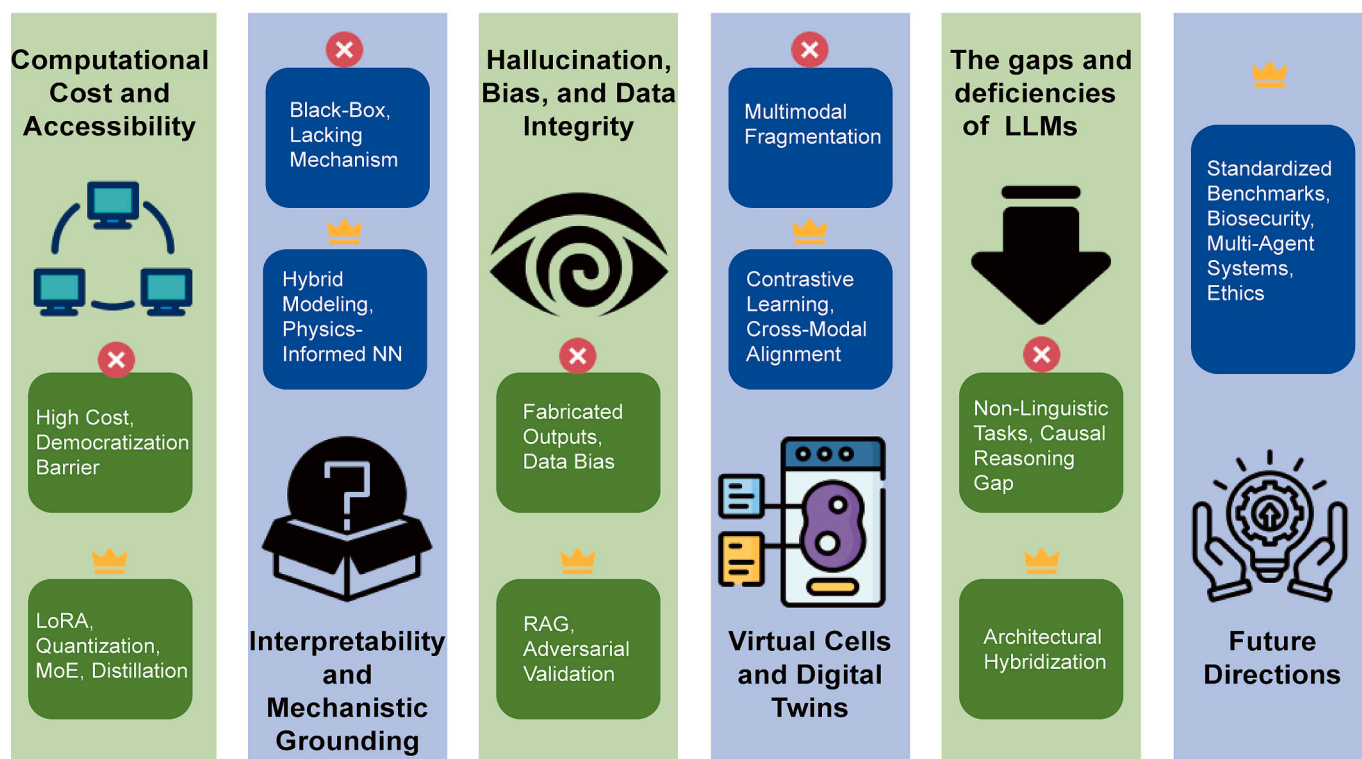


Fig. 3. Current challenges and future directions.

7.4. Virtual cells and digital twins

The ultimate test for computational synthetic biology is the creation of a predictive virtual cell, a digital twin that can simulate the emergent behavior of a living system from genetic code to metabolic phenotype (Jiang et al., 2025a; Qian et al., 2025b). Current LLMs excel within a single modality. The grand challenge is multimodal fusion: creating a unified model that seamlessly relates a DNA sequence, the 3D protein structures it encodes, the metabolic network these proteins form, and the dynamic physiology that results.

Addressing this requires contrastive cross-modal pre-training. Models must learn a shared latent representation where a gene, its protein product's structure, and its metabolic reaction are close neighbors in embedding space. Advances in models like AlphaFold3 for structure-complex prediction (Hennig, 2025) and genomic-scale models like Evo2 (Brixi et al., 2025) are foundational steps. The future lies in architectures that can perform “cross-modal reasoning,” for example, inferring the impact of a regulatory non-coding RNA (sequence) on the flux through a pathway (kinetics) by understanding its effect on transcription factor binding (structure).

7.5. The gaps and deficiencies of LLMs in synthetic biology

Despite the LLM revolution, established ML methodologies, which have been refined through decades of domain-specific optimization, continue to dominate problems anchored in non-linguistic representations. For instance, in the prediction of small-molecule pharmacokinetics and toxicity (ADMET), the governing principles are quantum mechanical and stereochemical (Kim et al., 2024). These nuances, essential for binding affinity and metabolic stability, are often “lost in translation” when reduced to textual SMILES strings. Consequently, graph neural networks (GNNs) (Monem et al., 2025) and classical QSAR models (Damiao et al., 2025) remain unchallenged, as their inductive biases are inherently aligned with the topological nature of molecules. Besides, the inference of emergent behaviors within microbial consortia relies on correlation-based ecological models and statistical network

analysis (Wang et al., 2024). The sequence-prediction logic of Transformers is fundamentally alien to the stochastic, non-linear interactions of species in a fluctuating environment. In future research, a hybrid architecture can be explored to address this issue: for instance, using GNNs to encode the geometric or topological state (e.g., molecular graphs), which is then projected into the LLMs' embedding space. This enables the LLMs to be responsible for reasoning about the function of the molecule, while the GNNs ensure the accuracy of its structural representation.

Conversely, in some areas, although there are some LLMs emerging, their effects are not significant. One example is the reasoning over long-range genomic regulation (Balakrishnan et al., 2025), where the combinatorial logic of enhancer-promoter communication across three-dimensional space defies the linear narratives that language models naturally construct. Perhaps most telling is the gap in autonomous experimental diagnosis. LLMs can identify a failed reaction (Qiu et al., 2025). However, determining the root cause and prescribing a recovery strategy requires causal reasoning beyond pattern recognition. This entails discerning whether the failure arose from an enzyme's kinetic limitation, an unanticipated metabolic burden, cryptic genetic instability, or another mechanistic factor. In these arenas, LLMs have demonstrated a provocative capacity to begin the conversation, but they lack the requisite vocabulary to finish it. Their current applications are thus best understood not as solutions, but as compelling, incomplete prototypes.

7.6. Future directions: validation, governance, and collaboration

The trajectory points toward specialized multi-agent ecosystems. No single model will master all of biology. Instead, we will see collaborations between a biological sequence design agent, a metabolic flux agent, a protocol generation agent, and a laboratory hardware agent, orchestrated by a master orchestrator agent that translates high-level goals into coordinated action plans. This mirrors the specialization of a human research team. However, such coordination is not without failure modes: agents may produce conflicting recommendations,

propagate errors across modules, or enter deadlock states where no consensus emerges. Addressing these risks requires built-in conflict resolution protocols, fallback strategies, and clear escalation pathways to human supervisors.

Equally critical is the distinction between computational tool autonomy and physical tool autonomy. While current agents can invoke databases or run simulations, true end-to-end autonomy demands integration with laboratory hardware. On this frontier, platforms like PLMeAE have made initial strides but remain reliant on human-defined protocols. Future systems must bridge this gap by developing standardized application programming interfaces (APIs) for robotic equipment and real-time data streams.

It is important to clarify what “autonomy” means in current synthetic biology implementations. Systems such as EVOLVEpro, PLMeAE, and SAMPLE are best characterized as operating in a human-on-the-loop rather than a human-out-of-the-loop paradigm. While they automate specific sub-tasks, such as variant proposal, protocol execution, and data integration, humans remain essential for defining objectives, curating initial training data, setting experimental boundaries, and validating final outcomes. Fully autonomous systems capable of making unsupervised critical decisions do not yet exist. Achieving such capabilities still needs technical advances in multi-agent coordination and hardware integration.

This future must be built on a foundation of rigorous validation and ethical governance. The field needs standardized benchmarks beyond narrow task performance, focusing on generalization to novel biological spaces and robustness to noisy, real-world data. Ethically, as the power to design life accelerates, robust frameworks for biosecurity and privacy, like using federated learning for sensitive data (Wen et al., 2023), and dual-use oversight are essential (Gillum, 2025; Li et al., 2025; Yao et al., 2024). The goal is not just to create powerful tools, but to build a responsible and collaborative partnership between human creativity and machine intelligence, steering synthetic biology toward its most beneficial and profound outcomes.

8. Conclusion

LLMs are transitioning from tools to discovery partners in synthetic biology. This marks an ontological shift: we are moving beyond applying computation to biology toward instilling biological understanding into the architectures themselves. By treating sequences as syntax, pathways as logic, and protocols as executable narratives, LLMs establish a unified semantic framework for translating and composing biological languages. This unification is dismantling key barriers. In biological sequence design, we advance from exploiting natural variation to *de novo* generation of functional code. In metabolic engineering, LLMs evolve into systemic reasoners that balance kinetics, regulation, and production. Most profoundly, in experimentation, they become the core of SDLs, shifting the rate-limiting step from human ideation to robotic throughput. However, achieving autonomous biological intelligence requires resolving inherent tensions. The efficiency-interpretability-safety trilemma demands scalable yet accessible and transparent models. Hallucinations reflect a deeper epistemological gap between statistical correlation and causal mechanism—a gap that cannot be closed by data alone but requires integrating biophysical constraints into learning itself. Notably, certain synthetic biology problems remain better suited to classical ML approaches, where LLMs are not yet the optimal solution.

Therefore, the path forward is conceptual as much as technical. It requires viewing LLMs not as oracles but as proposal engines within a refined human-machine dialogue. The future lies in specialized multi-agent systems, where autonomous domain-specific agents leverage their expertise, and their orchestrated collaboration is optimized to achieve superior global objectives.

CRedit authorship contribution statement

Anqiang Ye: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Bing-Ying Wang:** Formal analysis, Data curation. **Zhenshun Cheng:** Methodology, Investigation. **Changjiang Dong:** Supervision, Funding acquisition. **Feng-Biao Guo:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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